Do **not** write solutions on this page.

Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

9. [Maximum mark: 15]

Consider the function defined by $f(x) = \frac{1}{2}x^2 + kx + 13$, where $x \in \mathbb{R}$ and $k \in \mathbb{Z}^+$.

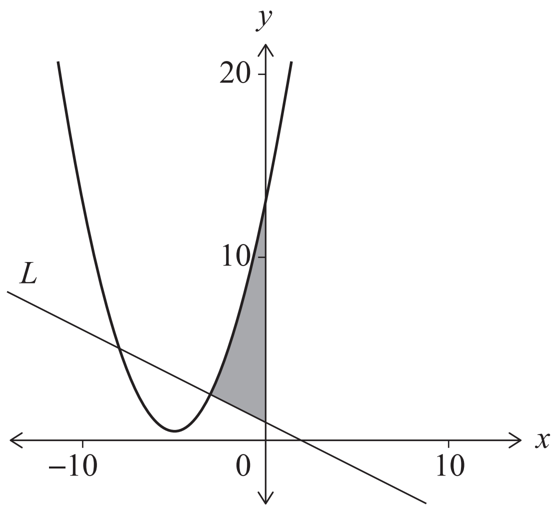
(a) Given that the equation $f(x) = 0$ has no real roots, show that the greatest possible value of k is 5. [2]

For the remainder of this question, consider the case $k = 5$.

(b) (i) Write down the equation of the axis of symmetry of the graph of f.

(ii) Hence, or otherwise, determine the coordinates of the minimum point on the graph of f. [3]

The following diagram shows the graph of f and a line L which is normal to the curve at $x = -3$. The shaded area shown is bounded by the curve, the line L and the y-axis.



(c) Show that the equation of L is given by $y = -\frac{1}{2}x + 1$. [5]

(d) Hence, find the shaded area. [5]